

TIANZE XU

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🎓 EDUCATION

Shanghai Jiao Tong University, School of Computer Science and Engineering 2025 – 2030

Ph.D. Student in Electronic Information (Artificial Intelligence)

- **GPA:** 3.69/4.0 **Advisor:** Prof. Pengfei Liu

Beijing University of Posts and Telecommunications, International School 2021 – 2025

Bachelor's Degree in E-Commerce and Law

- **GPA:** 91.7/100 (3.8/4.0) **Rank:** 1/165
- **Honors:** National Scholarship (2021); University Second-Class Scholarship and Merit Student (2022)

📖 PUBLICATIONS AND RESEARCH

ResearcherBench: Evaluating Deep AI Research Systems on the Frontiers of Scientific Inquiry

Co-first author (listed first) **Accepted at COLM 2026** arXiv:2507.16280 July 2025

- Introduced the first benchmark for evaluating Deep AI Research Systems (DARS) on frontier AI research: **65 questions** from authentic research settings spanning **35 AI subjects and 3 task types**, paired with expert-rubric insight evaluation and citation faithfulness/groundedness assessment.
- Led research formulation, dataset construction, factuality evaluation, and manuscript writing; evaluated representative systems and found that leading DARS reached **69–70%** insight coverage, outperforming other DARS by **20–30 points**, while exposing their open-consulting strength and high-faithfulness/low-groundedness pattern. Released the dataset and code.

Rubrics to Tokens: Bridging Response-level Rubrics and Token-level Rewards in Instruction Following Tasks

First author **Accepted at COLM 2026** arXiv:2604.02795 April 2026

- Proposed **Rubrics to Tokens (RTT)**, using a Token-Level Relevance Discriminator to project response-level rubric scores into fine-grained token rewards; designed RTT-GRPO to jointly optimize response- and token-level advantages and Intra-sample Token Group Normalization to remove length bias in multidimensional reward spaces.
- Led the method design and implementation; built a **62,714-example** token-level annotation corpus with a global/local × positive/negative taxonomy, minimal-rewrite and constraint-omission negatives, and a discriminator at **92.7%** token accuracy; conducted distributed RL training, ablations, and cross-model evaluation on ROLL. RTT achieved the best overall results across four instruction-following benchmarks, improving over the strongest RL baselines by **2.50 / 1.64 points** in instruction-/rubric-level accuracy with only **8.5%** additional training overhead. Open-sourced the full framework.

🏢 INDUSTRY EXPERIENCE

Alibaba Group, Taobao and Tmall Group, Business Technology Team Oct. 2025 – Sep. 2026

Large Language Model Algorithm Intern

- Conducted Agent-RL research on fine-grained reward modeling and credit assignment for complex instruction following; advanced the problem into the RTT method and a reproducible experimental pipeline, resulting in a first-author paper accepted at **COLM 2026**.
- Built an end-to-end distributed training pipeline on ROLL spanning data construction, token-level discriminator training/inference, advantage estimation, policy updates, and automated evaluation; supported multiple Qwen and Llama configurations and validated each component through systematic ablations and stability analysis.
- Drove RTT's rollout into the e-commerce data-analysis Agent loop: without touching existing tool protocols or execution chains, built a hybrid reward (rule signals + expert Rubrics + LLM-as-Judge) localized onto key content tokens, reasoning steps, and tool behaviors, training the Planner with GRPO/GSPO to jointly optimize overall outcome constraints and fine-grained step correction.

Lingxin Intelligence, Algorithm Intern May – Sep. 2023

- Contributed to **CharacterGLM**, an empathetic persona-consistent model released jointly with Tsinghua's CoAI Lab: built a large-scale distillation annotation pipeline over frontier LLMs (ChatGPT, Claude, etc.) to synthesize, clean, and label personalized dialogue data for model training.

🏆 AWARDS

- **First Prize**, Beijing Division, 13th and 14th Lanqiao Cup (Software, Group A), Apr. 2022 / 2023
- **First Prize**, Beijing Division, China International College Students' Innovation Competition (Project: "Painting Music with Colors"—an LLM-based multimodal music generation system), July 2024